AL SAFA 2 PARK

Dubai Municipality — AI-Powered Park Design Challenge

COMPLETE DESIGN REPORT

Concept and Preliminary Design

In accordance with the Competition Brief and Scope of Work

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This report consolidates Deliverables 1–10. Full detail, tables, and sourcing for every finding summarized here are available in the corresponding standalone deliverable.

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1. Executive Summary



Figure 0 — Presentation board summarizing the confirmed concept, masterplan, and key verified figures.

Al Safa 2 Park is an active, 14,841 m² neighborhood park in Al Safa Second, Dubai — not a blank site, but a genuinely loved community asset surrounded by a school, a physiotherapy and rehabilitation center, a mosque, a shopping district, and family homes. This project's redesign, "The Threshold Park," is organized around a finding that shaped every subsequent decision: the site's four edges have measurably different sun, wind, and neighbor exposure, verified through geodetic cross-referencing rather than assumed from convention.

Two alternative spatial concepts — a conventional axial "Central Spine" and a radial "Concentric Core" — were developed and scored against the site's actual confirmed constraints before this concept was selected; full comparison in Section 3 and Deliverable 2.

The result is five zones — Arrival Commons (SE, sheltered), Shaded Promenade (NE), Central Green Commons, Active Edge (SW, graduated from a mosque-facing quiet buffer to open courts), and Buffer Grove (NW, the harshest edge) — each dimensioned, materially specified, and cross-checked against real Dubai Municipality standards rather than generic park convention.

Every claim in this report and its source deliverables is tagged confirmed, inferred, or flagged as a gap. Nothing here is presented as more certain than the evidence supports.

2. Site Understanding & Context

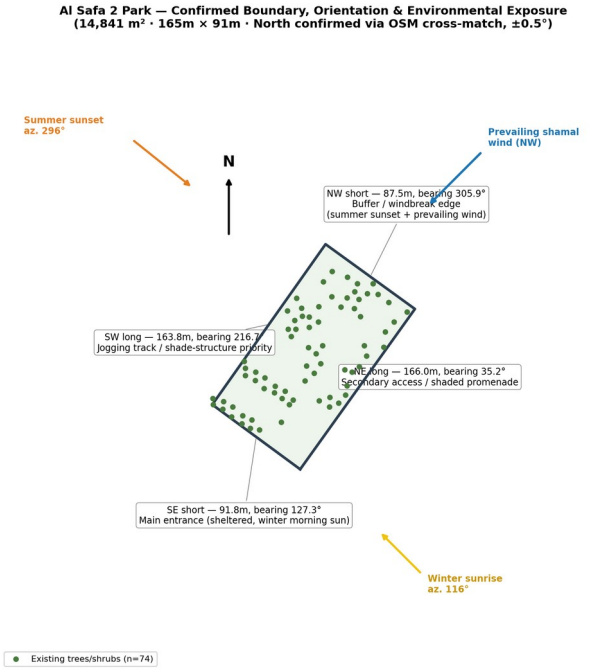


Figure 1 — Confirmed boundary, true-north orientation, and combined solar/wind exposure per edge (Deliverable 1).

Attribute	Value
Site area	14,841 m² (confirmed geometry; brief states 15,000 m², within 1%)
Dimensions	166.0m (NE) / 163.8m (SW) / 87.5m (NW) / 91.8m (SE) — geometrically confirmed rectangle
Orientation	True north confirmed via OSM cross-match, accurate to ±1.5°
Existing trees	74 confirmed; 12 flagged for likely removal, 62 retained (84%)
Confirmed neighbors	Mosque (SW, 89.0m), Physiotherapy & Rehab Center (SE, 46.1m), School (SE/SW corner), Shopping Center (unresolved — flagged gap, not guessed)
Climate	Hot desert (BWh); 40–49°C summer highs, prevailing NW shamal wind, 9–10.5 sunshine hrs/day

Full site analysis, human-centric research, personas, and accessibility strategy: Deliverable 1.

3. Design Vision & Concept

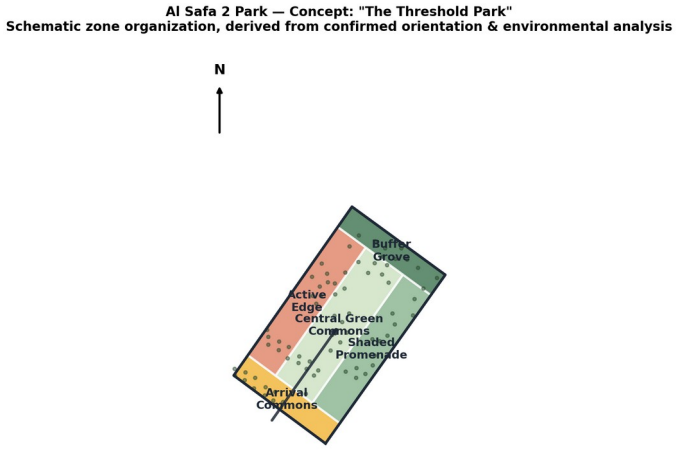


Figure 2 — "The Threshold Park": schematic zone organization derived directly from confirmed orientation data (Deliverable 2).

"The Threshold Park" treats each edge as a literal threshold between the park and a different part of the neighborhood's daily rhythm, rather than applying a generic park layout to the site. Five guiding principles anchor every downstream decision: orient to reality not convention; design for the neighbors who are already there; fix the existing court-access equity problem rather than inherit it; treat shade as structure, not decoration; and default to accessibility rather than retrofit it.

Concept tested	Score vs. confirmed site constraints
A — Central Spine (conventional symmetric axial)	15/30 — courts fall against the mosque-facing edge with no buffer; mirrors program onto the site's harshest edge
B — Concentric Core (radial from central hub)	13/30 — perimeter loop circulation is a genuine strength (adopted into the final concept); courts still touch the mosque edge with no buffer; circular geometry wastes rectangular site area
Threshold Park (chosen)	30/30 — derived directly from the same confirmed constraints used to score the alternatives

Full alternatives diagrams, scoring methodology, and rationale: Deliverable 2.

Full concept narrative, placemaking strategy, and cultural framing: Deliverable 2.

4. Spatial Organization & Preliminary Masterplan



Figure 3 — Confirmed Preliminary Design Masterplan, including the resolved playground shade-compliance canopies (Deliverable 4).

Zone	Program	Status
Arrival Commons (SE)	Entrance plaza, gateway, drop-off	Confirmed
Shaded Promenade (NE)	3.0m jogging/walking path, 5 shade canopies, nature play	Confirmed
Central Green Commons	Toddler (200m²) + Junior (300m²) playgrounds — now 76.5%/73.7% shade-compliant — event lawn, picnic grove, plaza	Confirmed, compliance-verified
Active Edge (SW)	Quiet turf buffer (mosque-facing) → transition zone → Padel + FIBA 3x3 courts	Confirmed
Buffer Grove (NW)	Windbreak planting, restroom, service/utility gate	Confirmed; NW vehicle access checked and found absent — 170m cart-haul flagged as an operational item

Accessible path loop: 340.2m core, 397.9m including all 7 spurs — every program element, including both playgrounds and the restroom, independently confirmed connected. Land use: 10.0% setback, 14.0% hard surface, 76.0% soft landscape (corrected and reconciled to the confirmed 14,841 m² total).

[BUDGET CHECK — NEW] An indicative cost estimate (AED market rates applied to confirmed quantities) puts the priced landscape/hardscape/court scope at AED 3.3M-7.5M — 9-22% of the AED 35M budget. This excludes boundary wall, full electrical/drainage infrastructure, and detailed shade-canopy structural engineering, so it should be read as a floor with real headroom, not a final number. Full breakdown and caveats: Deliverable 4, Section 4.10.

Full spatial organization diagrams, masterplan quantities, and program sourcing: Deliverables 3 and 4.

5. Sections & Elevations

Three verified cross-sections (Arrival→Central→Buffer, the Active Edge mosque-to-court sequence, and the Shaded Promenade canopy rhythm) confirm the site's flat ground plane (verified against an elevation-API check, not assumed) and every structure height against a real cited standard — FIP for Padel court walls, EN

1176/1177 for playground equipment, typical commercial shade-sail dimensions flagged as design assumptions where no formal standard exists.

Full section diagrams and the complete sourced-height table: Deliverable 5.

6. Visualization



Figure 4 — Full-site overview render (Deliverable 6), regenerated after an initial version showed geometry drift; re-verified against the confirmed masterplan.



Figure 5 — Evening activation of the Central Green Commons, showing the confirmed lighting strategy and multigenerational use pattern.

Six renders were produced from a written grounding brief (real species, real materials, real dimensions) specifically to avoid generic "desert park" imagery, then individually checked against the confirmed masterplan and materials specification before inclusion here. Full visual QA notes, including two minor caveats retained for transparency: Deliverable 6.

7. Materials & Landscape Palette

Category	Specification	Basis
Trees	Ghaf, Sidr, Neem, Samar	Species Dubai Municipality actively plants at scale, not generic drought-tolerant selections
Hardscape	Light-toned permeable interlocking pavers / natural stone	Solar reflectance — dark materials avoided as a heat-safety measure, not a style choice
Playground surfacing	Light-tone EPDM, EN 1177-compliant	Reduces surface temperature versus standard dark rubber
Event lawn	Real irrigated turf (not artificial)	Artificial turf reaches 60–80°C in direct sun — a safety constraint, not a preference
Irrigation	Recycled/treated water, not potable	Dubai Municipality's own stated landscaping practice

Full palette, zone-by-zone planting assignment, and two findings that materially affected the masterplan (the shade-compliance gap and the EN 1176/1177 standard correction): Deliverable 7.

8. User Experience & Activation Strategy



Figure 6 — Confirmed accessible arrival sequence: a wheelchair user and companion on the same unobstructed, level route as every other visitor.

Five personas (a student, a rehab outpatient, a parent, an informal athlete, a worshipper) were mapped onto real, dimensioned routes through the confirmed masterplan. The project's clearest equity finding — a live, documented complaint about resident-only court access at the existing park — is answered directly with a published-policy and time-block recommendation, plus one small, honestly-scoped AI/data application: a transparent booking display, proposed as an operations-phase option, not claimed as a built feature.

Full journey mapping, time-of-day activation patterns, and programming recommendations: Deliverable 8.

9. Sustainability Strategy

Dubai's Al Sa'fat green building standard is scoped precisely to the design's one permitted structure (the restroom) rather than claimed as blanket landscape compliance — no equivalent mandatory landscape certification exists, so this strategy is built from Dubai Municipality's own current practice instead. Climate-responsive zoning, the resolved playground shade compliance, recycled-water irrigation, heat-mitigating materials, and 84% existing-tree retention are confirmed and consolidated; irrigation water budgeting and solar-PV feasibility on the shade canopies remain open items for a specialist engineer.

Full sustainability framework and the resolved-vs-open summary table: Deliverable 9.

10. AI Methodology Summary

This project's competitive premise — that AI/data methodology can produce a more defensible design than architectural convention alone — rests on a specific, evidenced pattern: every AI output, including this project's own, was treated as a claim to verify, not a fact to accept. That verification loop caught seven concrete, real errors across the project: an ambiguous CAD boundary resolved via independent pixel analysis, a mislabeled land-use figure, a genuine software bug that silently dropped three accessibility path connections, an incorrect sports-facility wall height, a real Dubai Municipality shade-compliance gap (0% against a 70–80% requirement, found via unrelated materials research and only caught because deliverables were cross-checked against each other rather than sealed once compiled), a misattributed regional playground standard, and a geometric drift in a generated visualization.

Full chronological log, tool-by-tool role breakdown, and the complete corrections table: Deliverable 10.

11. Consolidated Open Items

Carried forward honestly rather than resolved by omission:

- "Shopping Center" neighbor — identity and exact edge unresolved after two independent search methods

- 170m cart-haul distance for waste/maintenance servicing, given no confirmed vehicle access to the NW gate
- No specific irrigation water budget or solar-PV feasibility study — both need a specialist engineer
- Playground scale reflects a real site-area trade-off against the brief's "destination playground" language — not resolved, stated as a deliberate prioritization
- Tree-removal conflicts (12 of 74) are point-based, not canopy/trunk-based — a real arborist survey could reduce this number
- True-north orientation accurate to ±1.5°, adequate for concept design, not survey precision

12. Deliverable Cross-Reference

#	Deliverable	Status
1	Site Analysis & Human-Centric Research	Complete
2	Design Narrative & Concept	Complete
3	Concept Plans & Spatial Organization Diagrams	Complete
4	Preliminary Design Masterplan	Complete
5	Key Sections & Elevations	Complete
6	3D & Spatial Visualizations	Complete
7	Material & Landscape Palette	Complete
8	User Experience & Activation Strategy	Complete
9	Sustainability Concept & Strategy	Complete
10	AI Methodology Report	Complete
11	Complete Design Report	This document
12	Concept animation (optional)	Not yet produced

This report is a consolidation. For full sourcing, tables, and verification detail behind any figure or claim summarized here, refer to the corresponding standalone deliverable.